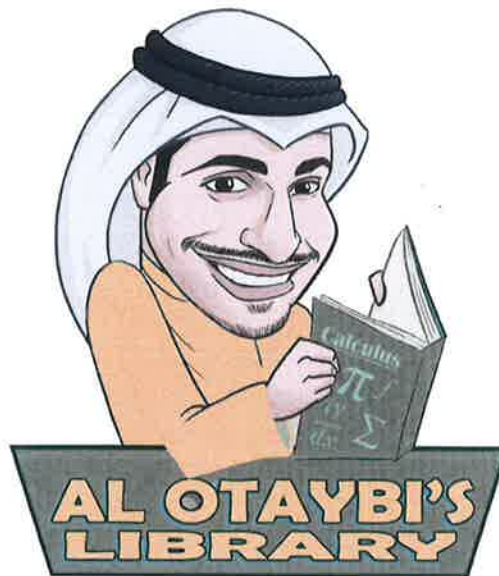


Kuwait University

2021/2022

Spring course



Calculus B

Final

Section : 6.2

6.2 Volumes

(15 b/b)

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Volume



خطوات الحل

$$x = x$$

$$y = y$$

(1) إيجاد نقاط التقاطع.

أعرفها

جدول لكل دالة متغيرة

(2) الرسم.

(3) وضع محور الدوران ثم وضع **element** بشكل عمودي عليه.

(4) تطبيق القانون مع مراعاة **thickness** وحدود التكامل.

$$V = \pi \int_a^b [(OR)^2 - (IR)^2] \text{ or } \frac{dx}{dy}$$

The same



$$V = \int_a^b A(x) dx \quad \text{or} \quad V = \int_c^d A(y) dy$$



انتبه



يجب أن تكون كل الدوال بدلالة **thickness**

Setup an integral for the volume of the solid generated by revolving the region bounded by the graphs of $y = -x^2 + 4$ and $x = y - 2$ about the line $y = -1$

$$y = -x^2 + 4$$

$$x = y - 2$$

about $y = -1$

$$y = x + 2$$

$$y = y$$

$$-x^2 + 4 = x + 2$$

$$x^2 + x + 2 - 4 = 0$$

$$x^2 + x - 2 = 0$$

$$(x - 1)(x + 2) = 0$$

$$x = 1 \quad x = -2$$

$$y = -x^2 + 4$$

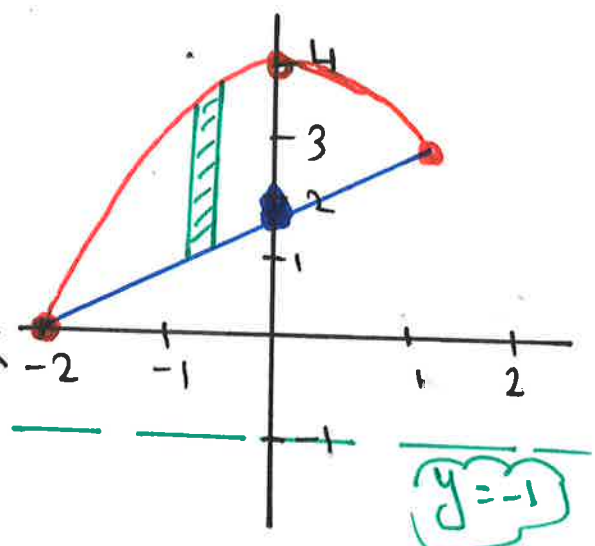
$$y = x + 2$$

x	-2	0	1
y	0	4	3

x	-2	0	1
y	0	2	3

$$V = \pi \int_{-2}^1 [(OR)^2 - (IR)^2] dx$$

$$= \pi \int_{-2}^1 [(-x^2 + 4 - (-1))^2 - (x + 2 - (-1))^2] dx$$



Let R be the region bounded by the graphs of

$$y + x^2 - 2x = 0 \text{ and } y + 2x = 0.$$

Set up an integral that can be used to find the volume of the solid generated by revolving R about the line $y = 1$.

$$y + x^2 - 2x = 0$$

$$y + 2x = 0$$

about $y = 1$

$$y = 2x - x^2$$

$$y = -2x$$

$$y = y$$

$$2x - x^2 = -2x$$

$$4x - x^2 = 0$$

$$x(4 - x) = 0$$

$$x = 0 \quad x = 4$$

$$y = 2x - x^2$$

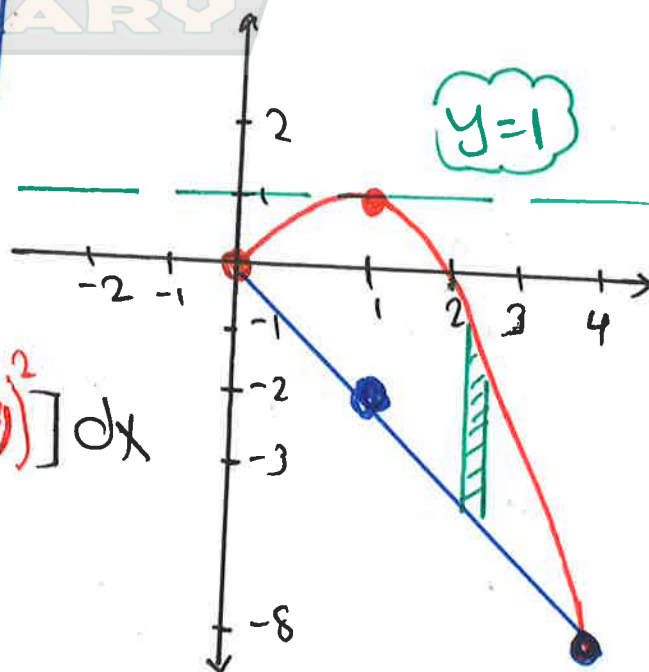
x	0	1	4
y	0	1	-8

$$y = -2x$$

x	0	1	4
y	0	-2	-8

$$V = \pi \int_0^4 [(OR)^2 - (IR)^2] dx$$

$$= \pi \int_0^4 [(1 - (-2x))^2 - (1 - (2x - x^2))^2] dx$$



Set up the volume of the solid obtained by rotating the region bounded by the given curves about the specified line

$$x = y^2, \quad x = 1 - y^2 \quad \text{about } x = 3$$

$$x = y^2 \quad x = 1 - y^2 \quad \text{about } x = 3$$

$$x = x$$

$$y^2 = 1 - y^2$$

$$2y^2 = 1$$

$$y^2 = \frac{1}{2}$$

$$y = \frac{1}{\sqrt{2}}$$

$$y = -\frac{1}{\sqrt{2}}$$

$$x = y^2$$

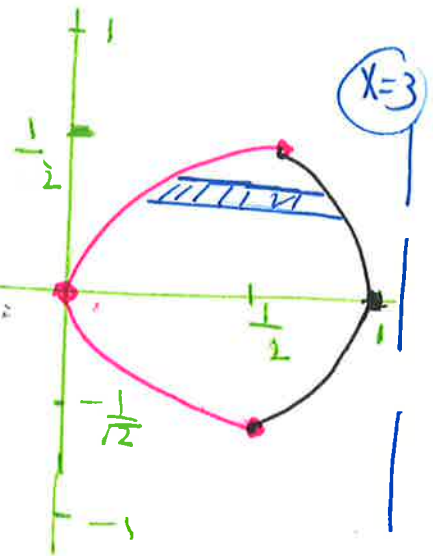
$$x = 1 - y^2$$

x	$\frac{1}{2}$	0	$\frac{1}{2}$
y	$\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

x	$\frac{1}{2}$	1	$\frac{1}{2}$
y	$-\frac{1}{\sqrt{2}}$	0	$\frac{1}{\sqrt{2}}$

$$V = \pi \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} [(OR)^2 - (ER)^2] dy$$

$$= \pi \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} [(3 - y^2)^2 - (3 - (1 - y^2))^2] dy$$



Set-up an integral for the volume of the solid generated by revolving the region bounded by $y = 2x - x^2$ and $y = 0$ about x -axis.

$$y = 2x - x^2 \quad y = 0 \quad \text{about } y = 0$$

$$y = y$$

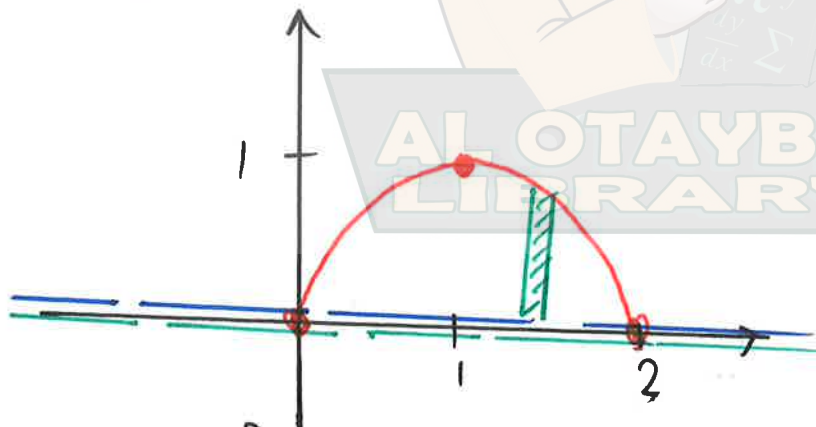
$$2x - x^2 = 0$$

$$x(2 - x) = 0$$

$$x = 0 \quad x = 2$$

$$y = 2x - x^2$$

x	0	1	2
y	0	1	0



$$\begin{aligned}
 V &= \pi \int_0^2 [(OR)^2 - (IR)^2] dx \\
 &= \pi \int_0^2 [(2x - x^2 - 0)^2 - (0 - 0)^2] dx
 \end{aligned}$$

Find the volume of the solid obtained by rotating

$$y = 2 - \frac{1}{2}x, y = 0, x = 1, x = 2, \text{ about } x - \text{axis.}$$

$$y = 2 - \frac{1}{2}x \quad y = 0 \quad x = 1 \quad x = 2 \quad \text{about } y = 0$$

$$y = y$$

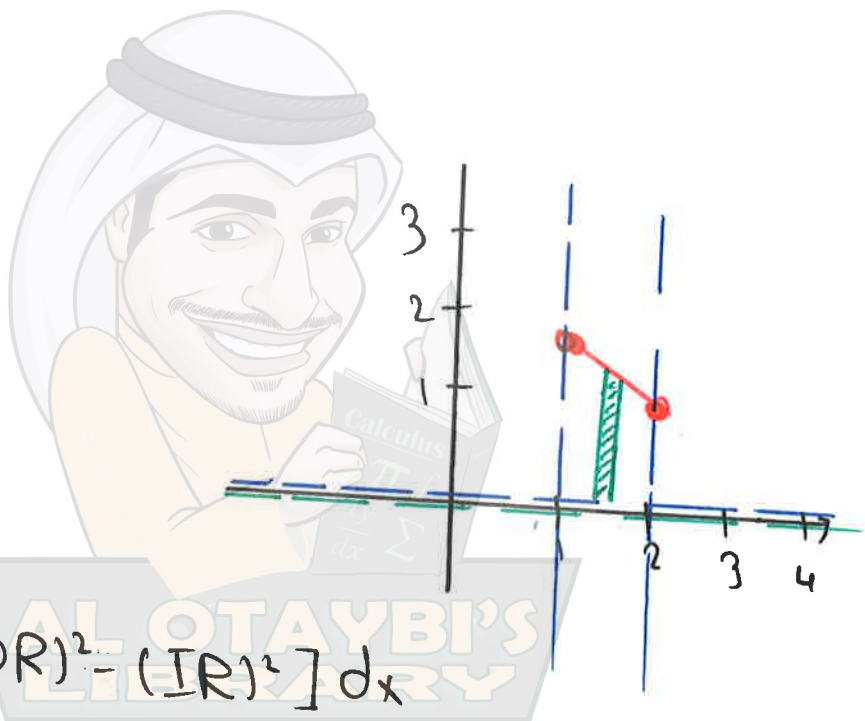
$$2 - \frac{1}{2}x = 0$$

$$2 = \frac{1}{2}x$$

$$x = 4$$

$$y = 2 - \frac{1}{2}x$$

x	1	2
y	$\frac{3}{2}$	1



$$\begin{aligned} V &= \pi \int_1^2 [(OR)^2 - (IR)^2] dx \\ &= \pi \int_1^2 [(2 - \frac{1}{2}x - 0)^2 - 0^2] dx \\ &= \pi \left[\frac{(2 - \frac{1}{2}x)^3}{(-\frac{1}{2}) 3} \right]_1^2 \\ &= \pi \left(-\frac{2}{3} \right) \left[(1)^3 - \left(\frac{3}{2} \right)^3 \right] \end{aligned}$$

Find the volume of the solid obtained by rotating the region bounded by $y = e^x$ and $y = 0$ for $0 \leq x \leq 1$, about the line $y = -1$. Fall 2017

$y = e^x$ $y = 0$ $0 \leq x \leq 1$ about $y = -1$

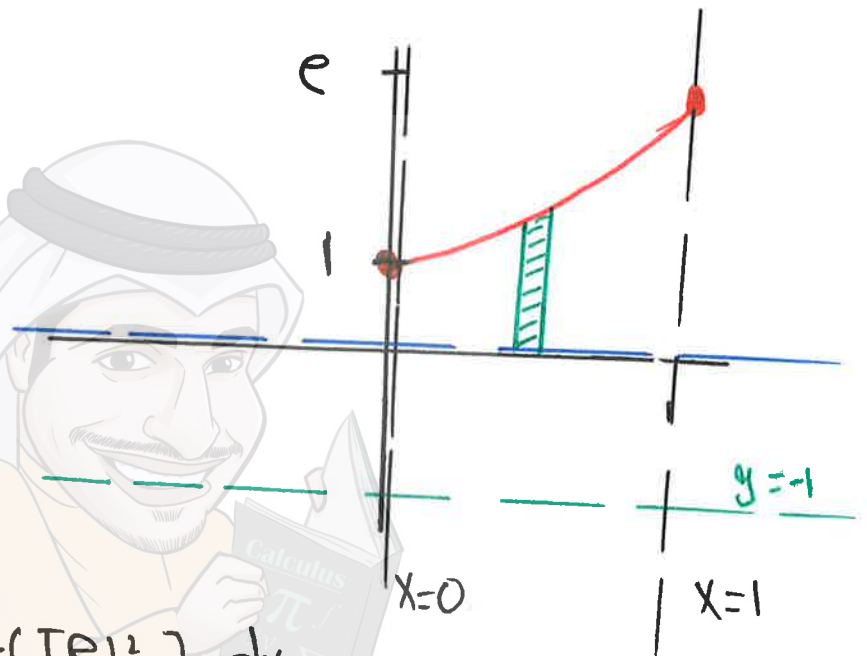
$y = y$

$e^x = 0$

None

$y = e^x$

x	0		1
y	1		e^1



$$\begin{aligned}
 V &= \pi \int_0^1 [(OR)^2 - (IR)^2] dx \\
 &= \pi \int_0^1 [(e^x - (-1))^2 - (0 - (-1))^2] dx \\
 &= \pi \int_0^1 [(e^x + 1)^2 - 1] dx \\
 &= \pi \int_0^1 [e^{2x} + 2e^x + 1 - 1] dx \\
 &= \pi \left[\frac{e^{2x}}{2} + 2e^x \right]_0^1 \\
 &= \pi \left[\frac{e^2}{2} + 2e - \left(\frac{1}{2} + 2 \right) \right]
 \end{aligned}$$

Set up the volume of the solid obtained by rotating the region bounded by the curves

$$x = y^2, \quad x = y^3 \quad \text{about} \quad a) x = 5 \quad b) x = -1$$

$$x = y^2$$

$$x = y^3$$

$$x = x$$

$$y^2 = y^3$$

$$y^2 - y^3 = 0$$

$$y^2(1 - y) = 0$$

$$y = 0$$

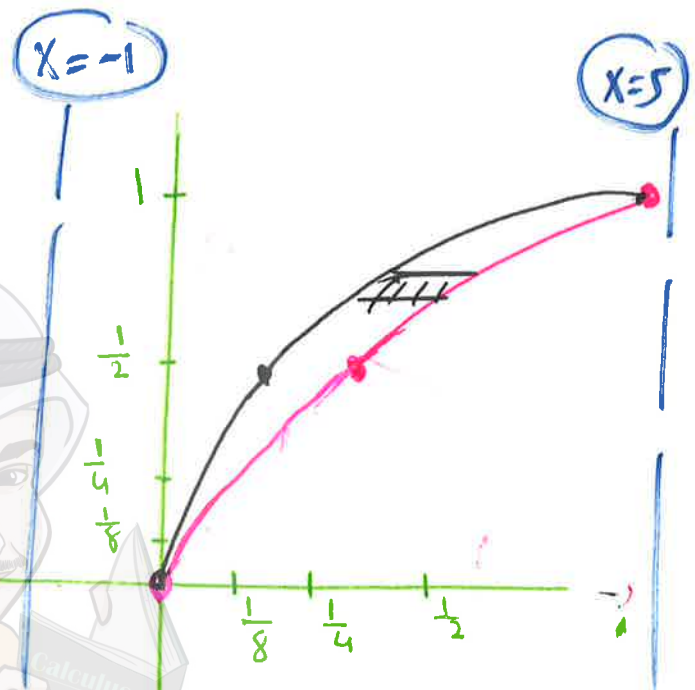
$$y = 1$$

$$x = y^2$$

$$x = y^3$$

x	0	$\frac{1}{4}$	1
y	0	$\frac{1}{2}$	1

x	0	$\frac{1}{8}$	1
y	0	$\frac{1}{2}$	1



(a) about $x = 5$

$$V = \pi \int_0^1 [(OR)^2 - (IR)^2] dy$$

$$= \pi \int_0^1 [(5 - y^3)^2 - (5 - y^2)^2] dy$$

(b) about

$$x = -1$$

$$V = \pi \int_0^1 [(OR)^2 - (IR)^2] dy$$

$$= \pi \int_0^1 [(y^2 - (-1))^2 - (y^3 - (-1))^2] dy$$

Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line

$y = \ln x$, $y = 1$, $y = 2$, $x = 0$ about the y - axis

$$y = \ln x$$

$$1 = \ln x$$

$$e^1 = e^{\ln x}$$

$$x = e^1$$

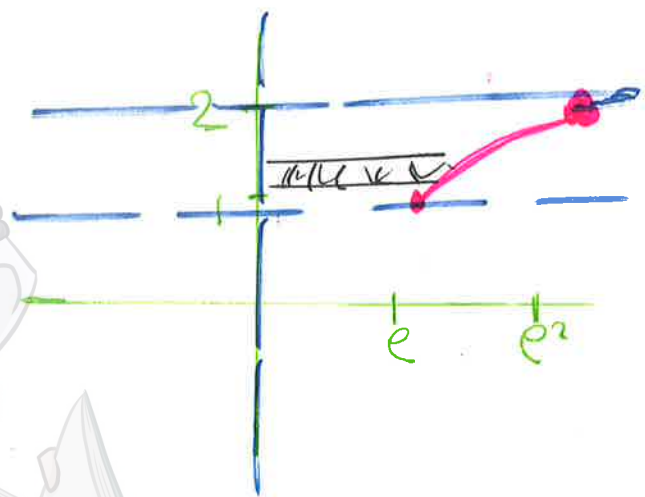
$$y = \ln x$$

$$2 = \ln x$$

$$e^2 = e^{\ln x}$$

$$x = e^2$$

$$x=0$$



$$y = \ln x \rightarrow x = e^y$$

x	e^1	e^2
y	1	2

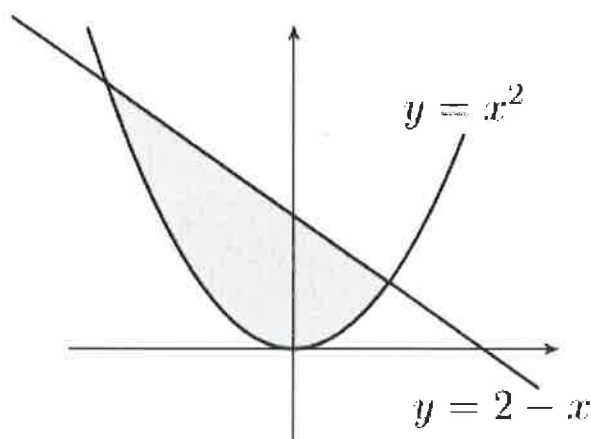
$$V = \pi \int_1^2 [(OR)^2 - (LR)^2] dy$$

$$= \pi \int_1^2 [(\cancel{\ln x} - 0)^2 - 0] dy$$

$$= \pi \int_1^2 (e^y - 0)^2 dy$$

$$= \pi \left[\frac{1}{2} e^{2y} \right]_1^2 = \frac{\pi}{2} (e^4 - e^2)$$

Set up (but not evaluate) an integral for the volume of the solid obtained by rotating the shaded region about the line $y = -1$.



Jan 2017

$y = x^2$ $y = 2 - x$ about $y = -1$

$y = y$

$x^2 = 2 - x$

$x^2 + x - 2 = 0$

$(x - 1)(x + 2) = 0$

$x = 1$ $x = -2$

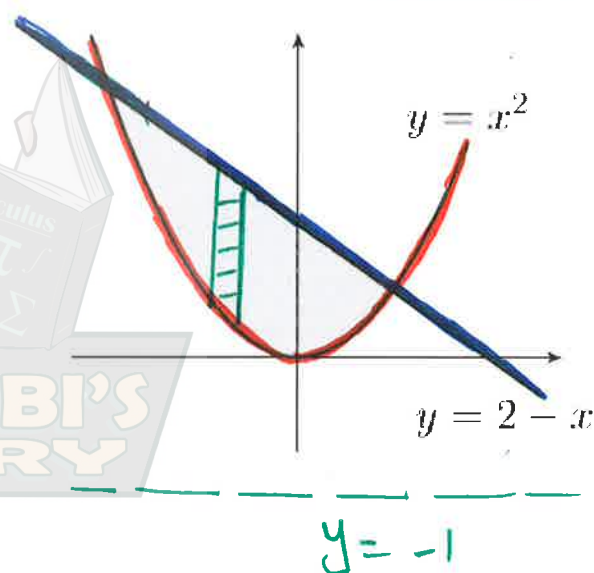
$y = x^2$

x	-2	0	1
y	4	0	1

$y = 2 - x$

x	-2	0	1
y	4	2	1

مطلبي الرسمة
أحد السؤال جيبني وكان فاعندي رسمة



$$V = \pi \int_{-2}^1 [(OR)^2 - (IR)^2] dx$$

$$= \pi \int_{-2}^1 [(2 - x - (-1))^2 - (x^2 - (-1))^2] dx$$

Set-up the volume of the solid obtained by rotating

$$y = x, \quad y = \sqrt{x} \quad \text{about } x = 2$$

$$y = x \quad y = \sqrt{x} \quad \text{about } x = 2$$

$$y = y$$

$$x = \sqrt{x}$$

$$x^2 = x$$

$$x^2 - x = 0$$

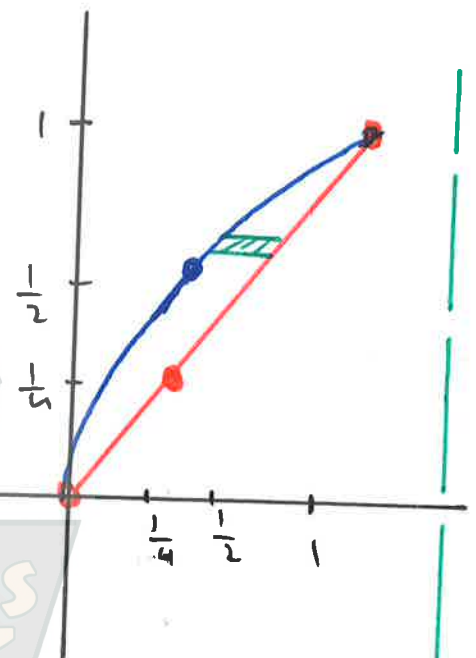
$$x(x - 1) = 0$$

$$x = 0 \quad x = 1$$

$$y = x \rightarrow x = y \quad y = \sqrt{x} \rightarrow x = y^2$$

x	0	1/4	1
y	0	1/2	1

x	0	1/4	1
y	0	1/2	1



$$V = \pi \int_0^1 [(OR)^2 - (IR)^2] dy$$

$$= \pi \int_0^1 [(2 - y^2)^2 - (2 - y)^2] dy$$

$x = 2$



Multiple choice



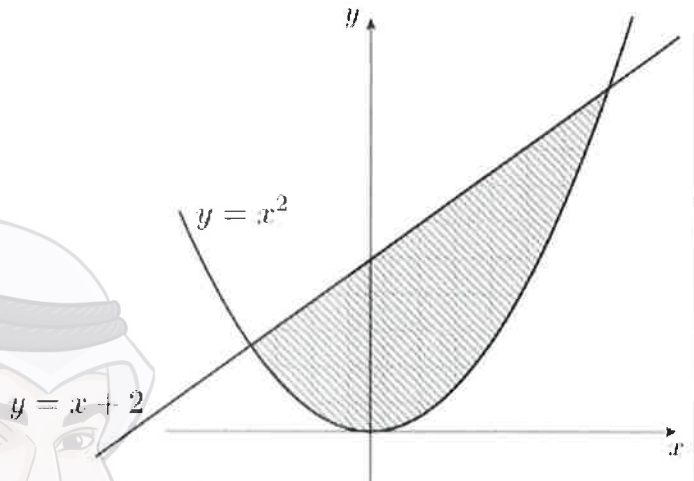
The shaded region is rotated about the line $y = -1$. Sketch the integral that represents the volume of the resulting solid.

(A) $\int_{-1}^2 \pi [(x+3)^2 - (x^2+1)^2] dx$

(b) $\int_{-1}^2 \pi [(x^2+1)^2 - (x+3)^2] dx$

(c) $\int_{-2}^1 \pi [(x+1)^2 - (x^2-1)^2] dx$

(d) $\int_{-2}^1 \pi [(x+3)^2 - (x^2+1)^2] dx$



May 2017

$y = x^2$ $y = x + 2$ about $y = -1$

$y = y$

$x^2 = x + 2$

$x^2 - x - 2 = 0$

$(x+1)(x-2) = 0$

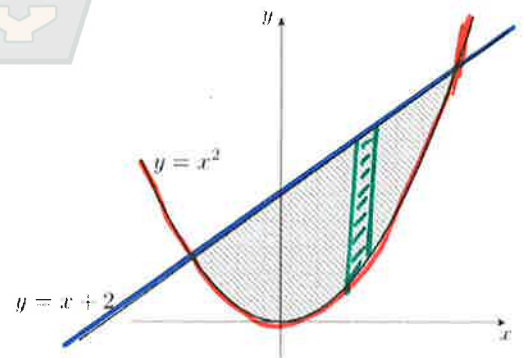
$x = -1$ $x = 2$

$y = x^2$

$y = x + 2$

x	-1	0	2
y	1	0	4

x	-1	0	2
y	1	2	4



$V = \pi \int_{-1}^2 [(OR)^2 - (IR)^2] dx$
 $= \pi \int_{-1}^2 [(x+2-(-1))^2 - (x^2-(-1))^2] dx$



Multiple choice



The volume of the solid obtained by rotating the region bounded by the curves $y = \sqrt{x}$ and $y = x^3$ about the vertical line $x = -2$ is

(A) $\int_0^1 \pi \left[(2 + y^2)^2 - (2 + \sqrt[3]{y})^2 \right] dy$

(b) $\int_0^1 \pi \left[(2 + \sqrt[3]{y})^2 - (2 + y^2)^2 \right] dy$

(c) $\int_{-2}^1 \pi \left[(2 + x^3)^2 - (2 + \sqrt{x})^2 \right] dx$

(d) $\int_{-2}^1 \pi \left[(2 + \sqrt{x})^2 - (2 + x^3)^2 \right] dx$

Fall 2018

$y = \sqrt{x}$

$y = x^3$

about $x = -2$

$y = y$

$\sqrt{x} = x^3$

$x = x^6$

$x - x^6 = 0$

$x(1 - x^5) = 0$

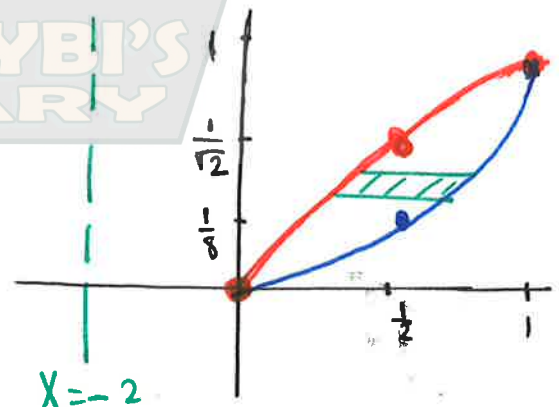
$x = 0 \quad x = 1$

$y = \sqrt{x} \rightarrow x = y^2$

$y = x^3 \rightarrow x = y^{\frac{1}{3}}$

x	0	$\frac{1}{2}$	1
y	0	$\frac{1}{\sqrt{2}}$	1

x	0	$\frac{1}{2}$	1
y	0	$\frac{1}{8}$	1



$$V = \pi \int_0^1 [(OR)^2 - (IR)^2] dy$$

$$= \pi \int_0^1 [(y^{\frac{1}{3}} - (-2))^2 - (y^2 - (-2))^2] dy$$

(b) —



Tricky question



Find the volume of the region obtained by revolving

$$y = x, y = 0, x = 2, x = 4 \text{ around } x = 1$$

$$y = x \quad y = 0 \quad x = 2 \quad x = 4 \quad \text{around } x = 1$$

$$y = y$$

$$x = 0$$

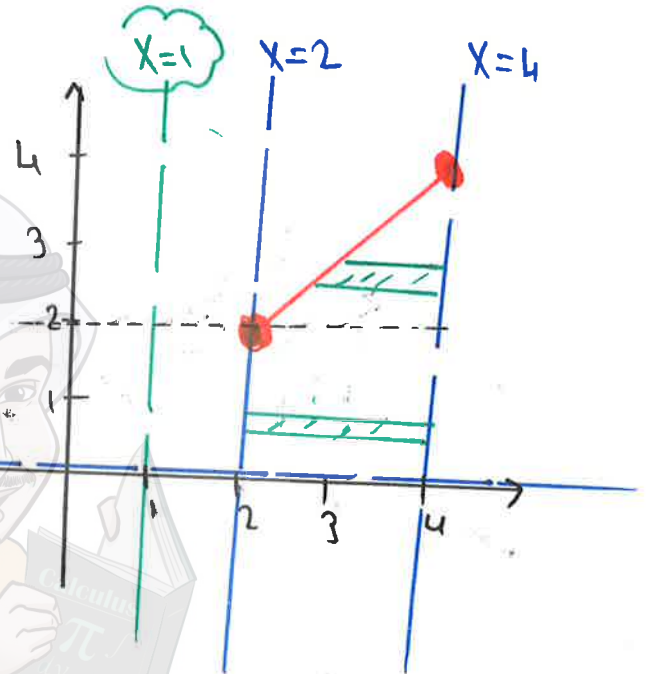
$$y = x \rightarrow x = y$$

x	2	4
y	2	4

$$V_T = V_1 + V_2$$

$$\begin{aligned} V_1 &= \pi \int_0^2 [(OR)^2 - (IR)^2] dy \\ &= \pi \int_0^2 [(4-1)^2 - (y-1)^2] dy \\ &= \pi \int_0^2 [9 - 1] dy \\ &= \pi \int_0^2 8 dy \\ &= \pi 8 [y]_0^2 \\ &= 8\pi [2-0] = 16\pi \end{aligned}$$

$$V_T = V_1 + V_2 = 16\pi + \frac{28}{3}\pi$$



$$\begin{aligned} V_2 &= \pi \int_2^4 [(OR)^2 - (IR)^2] dy \\ &= \pi \int_2^4 [(4-1)^2 - (y-1)^2] dy \\ &= \pi \left[9y - \frac{(y-1)^3}{3} \right]_2^4 \\ &= \pi \left[\left(36 - \frac{3^3}{3} \right) - \left(18 - \frac{1^3}{3} \right) \right] \\ &= \pi \left[36 - 9 - 18 + \frac{1}{3} \right] \\ &= \pi \left[9 + \frac{1}{3} \right] = \frac{28}{3}\pi \end{aligned}$$